Poster101K: Learning Poster-Specific Aesthetic Priors for Diffusion-based Poster Layout Generation (Supplementary Materials)

A Additional Dataset Analysis

A.1 Computation of Poster-Specific Statistics

To further support the rationale behind the proposed poster prior, we computed category-pair statistics for the Poster101K dataset. For each poster, we first aggregated the pairwise area ratio and pairwise IoU values by category, and then aggregated them across all posters in a poster-balanced manner. Specifically, for the area ratio analysis, we used the median of the logarithmic area ratios within each poster, and then took the median across all posters. For the overlap analysis, we calculate the average pairwise IoU for each pair of classes within each poster, and then take the average across all posters. Unless otherwise specified, the statistics for Poster101K include underlying elements, while those for Poster101K-NoUnderlay do not.

A.2 Color Aesthetics Evaluation

We evaluate the color aesthetics of Poster101K by using the Image Color Aesthetics Assessment (ICAA) method [1], trained initially on the ICAA17K dataset. Tab. 1 shows that Poster101K achieves a holistic color score of 5.54, close to ICAA17K's 5.88. This minor gap (0.34) may stem from different image types. ICAA17K mainly contains photographs of natural scenes, while images in our dataset are posters. This result demonstrates that Poster101K's color annotations can capture professionally designed aesthetics, providing a reliable basis for color-related research on poster layout generation.

Table 1: Comparison of color aesthetics assessment.

Dataset	Color Score	Temperature	Colorfulness	Color Harmony
ICAA17K[1]	5.88	5.90	5.56	7.12
Poster101K	5.54	5.16	3.64	5.43

A.3 Pairwise Area-Ratio Heatmap

Fig. 1 shows a heatmap of the pairwise log area ratios for Poster101K. Each cell reports the median poster balance of $log(S_i/S_j)$ between category pairs, where S_i and S_j denote the areas of the two elements, respectively. Positive values indicate that the row category tends to occupy a larger area than the column category. This heatmap suggests that poster layouts exhibit stable, category-level saliency patterns rather than arbitrary scale variations, supporting the use of visual hierarchy priors in the main model.

A.4 Category-Pair Overlap Heatmap

Fig. 2 shows a heatmap of category-pair overlaps on Poster101K. Each cell represents the average pairwise IoU value of poster alignment between two element categories. The results indicate that overlap in posters is not universally undesirable but rather category-dependent. In particular, overlaps involving background elements occur more frequently than many common foreground-foreground element pairs, which helps explain why completely suppressing overlap is too restrictive for poster layout generation. This observation further supports the poster-aware overlap modeling method investigated in this paper.

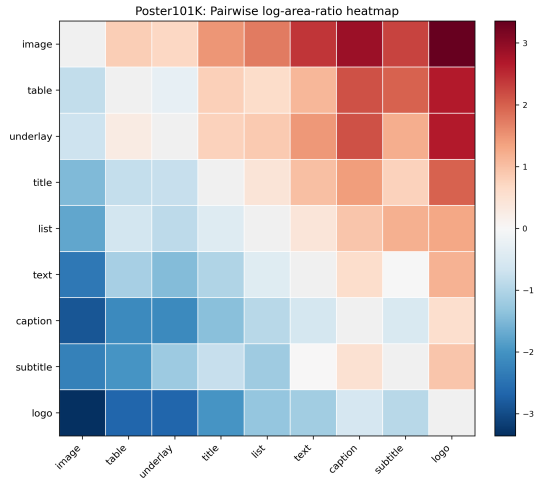

Figure 1: Pairwise Log Area Ratio Heatmap of Poster101K

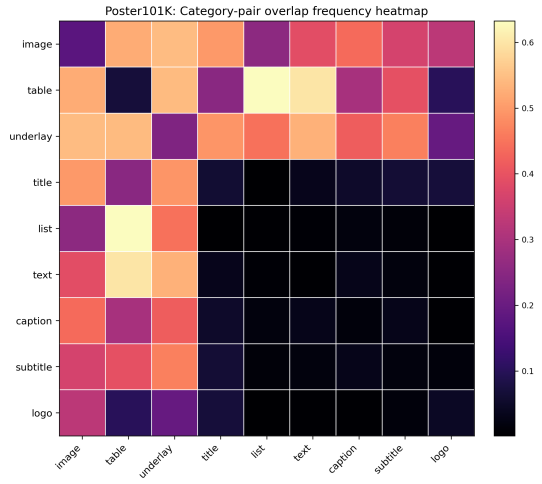

Figure 2: Category Pair Overlap Frequency Heatmap of Poster101K

Dataset	CGL				QBposter				Ours			
Task/Model	FID ↓	Max. ↑	Alg. ↓	Olp. ↓	FID ↓	Max. ↑	Alg. ↓	Olp. ↓	FID ↓	Max. ↑	Alg. ↓	Olp. ↓
U-Gen												
LayoutDM	17.8	-	0.65	70.4	12.6	-	0.46	85.4	9.61	-	0.73	49.8
LACE	15.7	-	0.51	**55.7**	7.71	-	**0.31**	71.3	7.89	-	0.63	**40.5**
Poster101K	**13.5**	-	**0.49**	65.1	**7.18**	-	0.38	**59.5**	**4.38**	-	0.64	55.1
C-Gen												
LayoutDM	18.8	0.32	0.60	54.3	9.24	0.12	0.39	79.3	8.97	0.22	0.67	52.8
LACE	11.1	0.34	**0.56**	50.6	7.96	0.14	**0.21**	**65.6**	7.41	0.22	0.56	**47.0**
Poster101K	11.3	**0.35**	0.59	49.5	**7.25**	0.14	0.35	71.4	**3.32**	**0.23**	**0.51**	58.9
CS-Gen												
LayoutDM	9.23	0.34	0.63	48.7	7.94	0.18	0.39	78.7	8.32	0.29	0.67	53.3
LACE	8.83	0.34	**0.56**	49.2	7.60	0.20	0.22	77.1	6.29	0.31	0.55	**46.2**
Poster101K	8.67	0.35	0.59	46.0	7.53	**0.20**	**0.21**	**74.9**	**2.24**	0.29	0.57	58.6
Completion												
LayoutDM	18.7	0.39	0.50	40.6	9.65	0.15	0.46	68.8	8.72	0.25	0.65	35.4
LACE	16.4	0.40	0.45	38.9	8.44	0.17	**0.27**	62.7	6.69	0.29	0.64	**26.3**
Poster101K	**15.9**	**0.41**	0.50	39.7	**8.25**	0.17	0.30	**52.8**	**4.46**	0.29	0.63	41.1
Refinement												
LayoutDM	8.83	0.42	0.48	52.9	7.60	0.17	0.41	75.4	5.29	0.39	0.51	51.7
LACE	**3.62**	0.46	0.36	48.8	7.39	0.21	**0.21**	**70.8**	3.32	0.44	0.42	40.3
Poster101K	3.80	0.46	0.42	47.5	**6.10**	**0.24**	0.29	71.1	**1.15**	**0.45**	**0.40**	53.2

Table 2: Quantitative results on three datasets for U-Gen, C-Gen, and CS-Gen. The best results are highlighted in bold.

A.5 Annotation and Preprocessing Details

Poster101K annotations are stored in XML format, derived from the raw percentage annotations exported by Label Studio. For each element, the XML file records the class label, a list of bounding box, and optional color palette attributes. During preprocessing, we converted these annotations into normalized, axis-aligned bounding boxes and class labels for use by the layout model. We also constructed Poster101K-NoUnderlay by removing the underlying annotations while keeping all other labels unchanged. This preprocessing workflow ensures consistency between dataset analysis and layout generation experiments.

B Extended Quantitative Results

B.1 Additional Public Benchmark Results

Our model demonstrates highly competitive performance across all layout generation tasks and other datasets, consistently achieving state-of-the-art results in key metrics while exhibiting particular strengths in FID and MaxIoU. As shown in Tab. 2, for Unconditional Generation (U-Gen), Ours achieves the best FID across all datasets (13.5 on CGL, 7.18 on QBposter, 4.38 on our dataset) and leads in Alignment on CGL (0.49), though LACE outperforms in Alignment and Overlap for QBposter. In Conditional Generation (C-Gen), our model records the highest MaxIoU on CGL (0.35) and our dataset (0.23), while achieving the best FID on QBposter (7.25) and our dataset (3.32). For CS-Gen, our model delivers the best FID on our dataset (2.24), the highest MaxIoU on QBposter (0.20), and top Alignment on QBposter (0.21), while maintaining strong Overlap on QBposter (74.9). In Completion tasks, our model dominates FID across all datasets (15.9 CGL, 8.25 QBposter, 4.46 our dataset) and MaxIoU on CGL (0.41), while for Refinement, it excels in FID on QBposter (6.10) and our dataset (1.15), MaxIoU on QBposter (0.24) and our dataset (0.45).

B.2 Transfer to the DLT Backbone

To verify that the proposed prior constraints are not tied to a specific diffusion backbone network, we integrated them into the DLT model [2] and present the results in Tab. 3. Similar to the trends observed in the original appendix experiments, the improved model (DLT*) outperforms the original DLT model in terms of generation fidelity and alignment across all three modes: U-Gen, C-Gen, and CS-Gen. These results indicate that the hierarchical structure and overlapping prior distribution capture transferable regularities in poster layouts, rather than implementation-specific heuristics.

Dataset		Ours			
Task	Model	FID ↓	pIOU ↓	Olp. ↓	Align. ↓
U-Gen	DLT	13.5	18.1	**50.7**	0.97
	DLT*	**6.0**	**17.1**	60.7	**0.67**
	w/o C_{OLC}	8.2	17.8	54.8	0.72
	w/o C_{VHC}	9.7	18.0	62.3	0.81
C-Gen	DLT	12.6	**20.8**	**56.7**	0.82
	DLT*	**4.5**	21.8	67.8	**0.66**
	w/o C_{OLC}	6.9	21.3	62.6	0.75
	w/o C_{VHC}	8.6	21.6	69.1	0.71
CS-Gen	DLT	8.4	22.2	**56.8**	0.66
	DLT*	**3.7**	**20.9**	63.6	**0.57**
	w/o C_{OLC}	4.4	21.5	64.5	0.61
	w/o C_{VHC}	5.6	21.8	61.8	0.59

Table 3: Ablation study on DLT. The best results are highlighted in bold.

C Implementation Details and Qualitative Results

C.1 Training Details

All models were implemented using PyTorch 1.9 and trained on four NVIDIA RTX 3090 GPUs. Consistent with the main paper, the diffusion process used 1,000 training time steps with linear noise scheduling; the inference process used 100 time steps of DDIM sampling. Training is conducted in two phases: first, the diffusion backbone network is optimized without introducing new priors; then, it is fine-tuned with hierarchical and overlap constraints enabled.

C.2 Results on U-Gen, C-Gen, and CS-Gen

Fig. 3-5 provide further qualitative comparisons of U-Gen, C-Gen, and CS-Gen. Compared to the best baseline, our methods tend to generate layouts with clearer structural highlights, more coherent grouping of elements, and more reasonable use of overlaps. These figures complement the main text, demonstrating that quantitative improvements are consistently reflected in the visual quality of the layouts.

References

[1] Shuai He, Anlong Ming, Yaqi Li, Jinyuan Sun, Shuntian Zheng, and Huadong Ma. 2023. Thinking Image Color Aesthetics Assessment: Models, Datasets and Benchmarks. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*. IEEE, 21781–21790.

[2] Elad Levi, Eli Brosh, Mykola Mykhailych, and Meir Perez. 2023. DLT: Conditioned layout generation with Joint Discrete-Continuous Diffusion Layout Transformer. In *IEEE/CVF International Conference on Computer Vision, ICCV, Paris, France*. 2106–2115. doi:10.1109/ICCV51070.2023.00201

Figure 3: Qualitative comparison between our method and LACE in the U-Gen task.

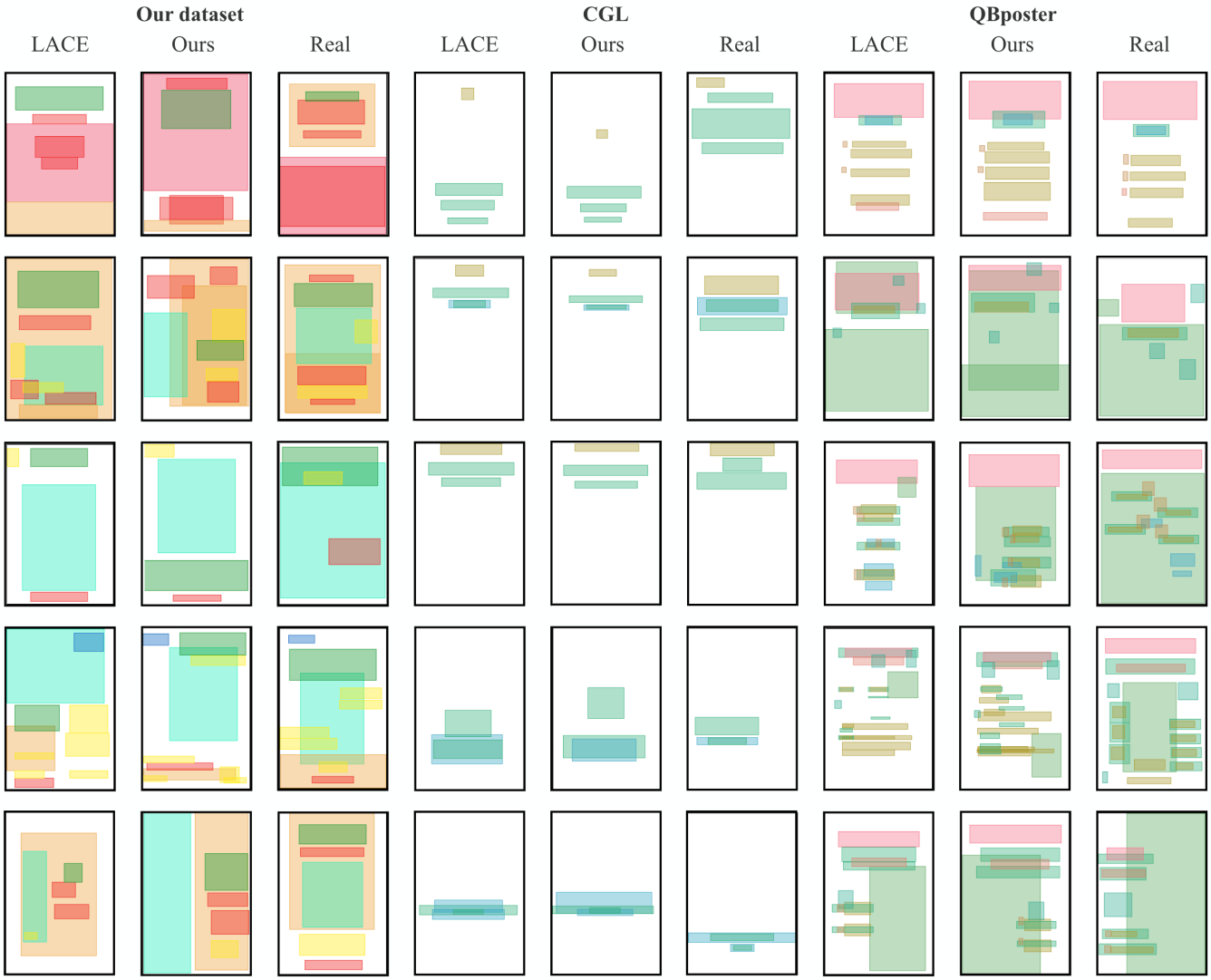

Figure 4: Qualitative comparison between our method and LACE in the C-Gen task.

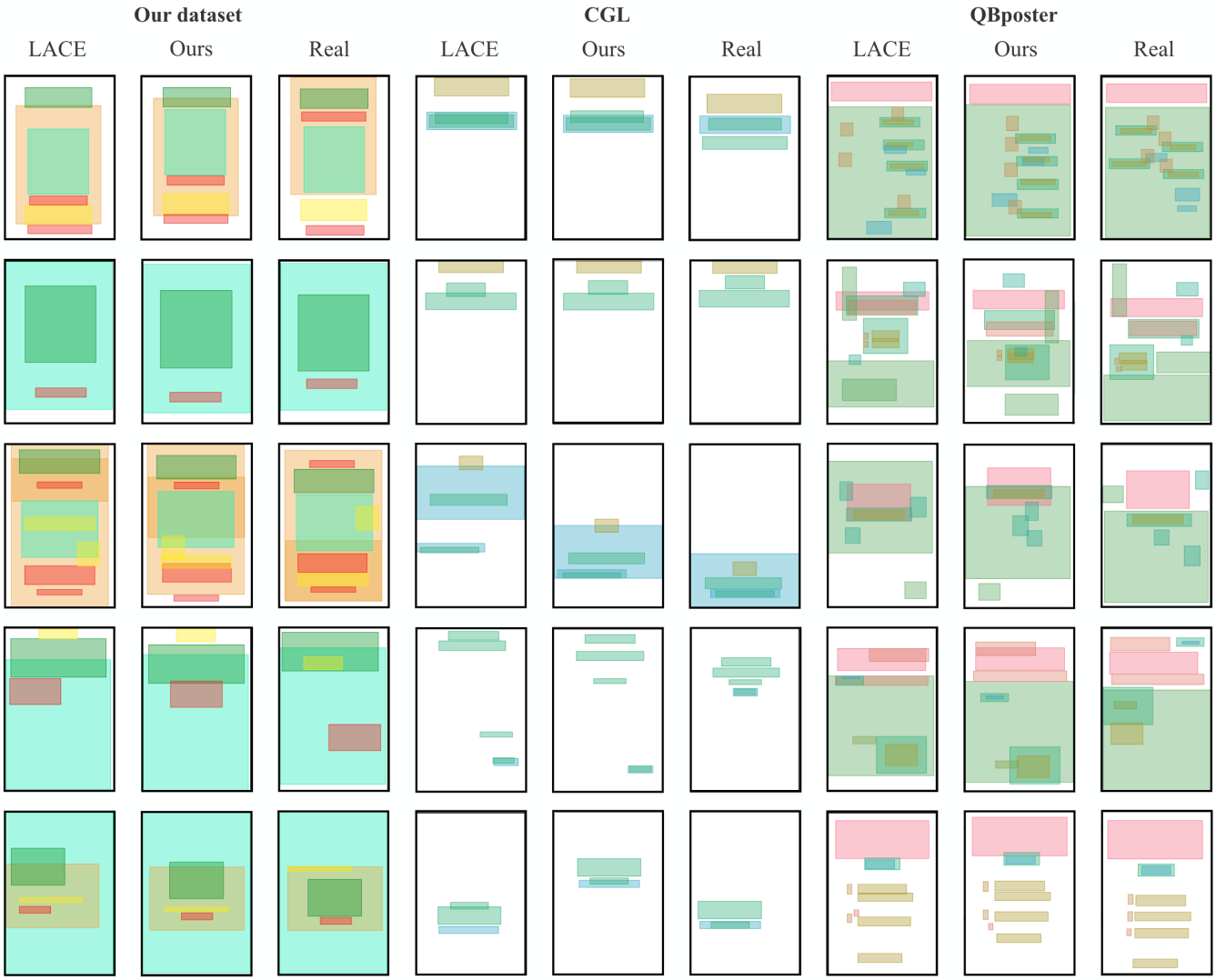

Figure 5: Qualitative comparison between our method and LACE in the CS-Gen task.